



CONFIDENTIAL

DISASTER RECOVERY PLAN

SAP ERP Central Server DR Plan

SAP ERP — Mission-critical SAP S/4HANA central instance running core business processes (FI/CO, MM, SD, PP, HC)

Service Owner

Finance & Operations IT

Criticality

CRITICAL

RTO Target

120 minutes (2h 0m)

RPO Target

30 minutes (0h 30m)

Plan ID

7769d7ac-1ba2-4121-8f38-a501cd0e987d

Status

DRAFT

Document Date

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Standard

ISO 22301:2019

PT Datacomm Diangraha

IT Service Resilience

Confidential — Internal Use Only

Disaster Recovery Plan

7769D7AC

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Plan Metadata

Field	Value
Service	SAP ERP
Service Owner	Finance & Operations IT
Criticality	critical
RTO Target	120 minutes (2h 0m)
RPO Target	30 minutes (0h 30m)
Plan ID	7769d7ac-1ba2-4121-8f38-a501cd0e987d
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Plan Quality Score

50/100 FAIR

Signal	Weight	Detail
✓ Substantive section content	+25	14/14 sections look substantive
✗ Plan approved	+20	Plan is not approved yet
✗ Section-level readiness	+15	0/14 sections approved
✗ Service and business owners	+10	Business owner or service owner missing
✓ RTO/RPO targets	+10	RTO 120m / RPO 30m
✓ Recovery strategy/context	+10	Recovery context exists
✗ Audit evidence linked	+5	0 evidence item(s)
✓ Recently maintained	+5	Updated within 180 days

Plan Sections

1. Context and Scope

ISO 22301:2019 Clause 4 — 151 words

Service: SAP S/4HANA Central Server (CI + HANA DB, 2 TB)

Service Owner: Finance & Operations IT

Criticality: Tier-1 / Critical

Audience: DR Coordinator, Incident Commander, SAP Basis team, Integration owners, Exec Crisis Team

In Scope

- Primary SAP S/4HANA Central Instance, HANA database, Fiori front-end
- 12 satellite integrations (banking, payroll, warehouse, e-invoicing, etc.)
- 3,000 internal named users (FI/CO, MM, SD, PP, HCM)
- DR site compute, storage replication, network failover

Out of Scope

- Local satellite systems' own DR (handled at integration level)
- End-user devices, printers, network access layer
- Non-SAP analytics/reporting warehouses (daily refresh acceptable)

Regulatory / Compliance Mapping

- ISO 22301:2019 Clauses 4-10
- SOX (financial reporting integrity — FI/CO)
- GDPR (HCM/payroll personal data residency)
- Local e-invoicing mandates, tax authority retention rules

Purpose: Define how the SAP S/4HANA Central Server is recovered within RTO 120 min / RPO 30 min following a disruption.

2. Leadership and Ownership

ISO 22301:2019 Clause 5 — 222 words

Role	Name / Function	Responsibility
Plan Owner	Head of Finance & Operations IT	Accountable for DR plan lifecycle, budget, sign-off
DR Coordinator	SAP Basis Lead	Activates plan, coordinates technical recovery
Incident Commander (IC)	IT Service Continuity Manager	Declares disaster, owns command, go/no-go
Approver	CIO + CFO joint sign-off	Authorises invocation and stand-down
SAP HANA DBA Lead	Senior DBA	Database recovery, replication integrity
Integration Lead	Middleware/PI Owner	Restores 12 satellite interfaces
Crisis Comms Lead	Corporate Communications	Stakeholder messaging cadence

RACI Matrix (Recovery Phases)

Activity	Plan Owner	DR Coord	IC	SAP Basis	Integration	Comms
Declare disaster	A	C	R	I	I	I
Activate failover	A	R	C	C	C	I
Verify services	A	R	A	R	R	I
Communicate to users	A	C	A	I	I	R
Stand-down	R	C	R	I	I	I

ISO 22301 §5.2, §5.3 — leadership and roles.

3. Planning and Objectives

ISO 22301:2019 Clause 6 — 150 words

Recovery Objectives

Metric	Target	Driver
RTO	120 minutes	Maximum tolerable downtime before FI/CO posting halt causes material financial impact
RPO	30 minutes	Last acceptable committed HANA transaction timestamp
MBO (Minimum Business Objective)	Restore FI/CO + payroll within RTO; MM/SD/PP within +60 min	

Recovery Success Criteria

1. HANA DB consistent and online at DR site
2. SAP application services (CI, ASCS, Fiori) started
3. All 12 satellite integration channels open and processing
4. Smoke test: one FI posting, one payroll record, one sales order round-trip succeeds
5. $\geq 95\%$ of 3,000 users able to log on within 60 min post-declare

Measurement

- Real recovery time vs RTO recorded in DR register
- Real data loss vs RPO measured from last replicated log timestamp
- Quarterly KPI reporting to Risk Committee

Aligned with ISO 22301 §6.2.

4. Support and Resources

ISO 22301:2019 Clause 7 — 134 words

Resources & Tools

- DR site (warm-standby, HANA System Replication + ASCS cluster replication)
- SAN replication (synchronous to metro, async to DR region)
- DNS failover automation (Route 53 / equivalent)
- SAP HANA Studio, SAP MMC, central monitoring (Solman)
- Pre-staged DR runbook binder + offline copies

Budget (annual)

- DR site hosting: €X
- Replication links: €Y
- 4 exercises: €Z
- 3rd-party DR advisor retainer: €W

Vendor Contracts

- Hyperscaler / co-lo DR site — 99.9% SLA, 4h restore of compute
- Storage replication vendor — RPO $\leq$ 5 min capability
- SAP Premium Engagement — escalation support

Communication Channels

- War-room bridge (Teams persistent channel)
- SMS tree via alerting tool (Everbridge/Similar)
- Status page (internal + customer-facing)
- Out-of-band phone tree (printed, sealed envelope)

Per ISO 22301 §7.1.

5. Operational Control

ISO 22301:2019 Clause 8.1 — 136 words

Day-to-Day Operational Controls (Pre-Event)

1. Replication Health

- HANA System Replication status checked every 5 min via Solman
- SAN replication lag alert threshold: > 60 sec → page on-call

1. Backup Cadence

- HANA full backup nightly 02:00, logs every 15 min
- Backups replicated to DR region; restore-test monthly on a non-prod clone

1. Monitoring

- 24/7 NOC watches SAP central services, ASCS, DB, integration middleware
- Synthetic transaction (FI posting) every 30 min from probe

1. Change Control

- All SAP transports gated by CAB; DR impact assessment required
- Configuration drift between PRD and DR detected weekly

1. Patch & Maintenance

- HANA revisions kept $\leq$ 1 quarter behind latest; DR site mirrored within 14 days

1. Documentation Currency

- Runbook reviewed monthly; topology diagram regenerated quarterly

Per ISO 22301 §8.1, §8.5.

6. Business Impact Analysis

ISO 22301:2019 Clause 8.2.2 — 236 words

Business Impact Analysis Summary

Process	Module	Users Affected	1h Impact	4h Impact	24h Impact	Tier
Financial posting / close	FI/CO	800	High — close blocked	Severe — audit risk	Critical — regulatory breach	**T1**
Payroll run	HCM + payroll int.	3,000 (indirect)	Medium	High — missed pay	Critical — legal	**T1**
Sales order entry	SD	600	Medium	High — revenue loss	High	**T1**
Procurement / goods receipt	MM	400	Low	Medium — ops impact	High	**T2**
Production planning	PP	350	Low	Medium	High — plant stop	**T2**
Warehouse / outbound	WM + int.	500	Medium	High — SLA breach	High	**T1**
Banking interface	FI/bank	800	High	Critical — liquidity	Critical	**T1**
E-invoicing (legal)	SD/FI	External	Low	High — non-compliance	Critical	**T1**

Impact Time Bands (per ISO 22301 §6.2)

- < 1h: tolerable for T2; rising concern for T1
- 1-4h: T1 enters escalation, customer comms activated
- > 24h: regulatory notification considered

Tiering result: 7 processes T1 → drive the 120 min RTO.

7. Risk Assessment

ISO 22301:2019 Clause 8.2.3 — 190 words

Top Threats (Likelihood × Impact)

Threat	Likelihood	Impact	Inherent Risk	Mitigation	Residual
Primary data centre outage	Medium	Severe	High	HANA SR + DR site warm	Low
HANA database corruption	Low	Severe	High	Daily backups + log replay, pre-flight checks	Medium
Cyberattack / ransomware	Medium	Severe	High	Immutable backups, MFA, network seg, IR plan	Medium
Replication link failure	Medium	High	High	Dual link, async fallback, monitoring	Low
Integration middleware failure	High	Medium	Medium	Redundant middleware, queue replay	Low
Cloud provider DR region failure	Low	Severe	Medium	Multi-AZ DR, quarterly failover proof	Low
Human error during recovery	Medium	High	High	Drills, two-person integrity, runbook automation	Medium
Loss of skilled SAP staff	Medium	High	Medium	Cross-training, external retainer	Medium

Reviewed quarterly; risk register linked to ERM.

ISO 22301 §6.1.

8. Recovery Strategy

ISO 22301:2019 Clause 8.3 — 156 words

Chosen Strategy: Warm-Standby with Pilot-Light for Integrations

Rationale

- RTO 120 min excludes backup-restore (HANA restore of 2 TB > 4 h) and cold-standby
- Hot-standby (active-active) deemed over-engineered for 3000 users with 99.9% availability target
- **SAP HANA System Replication** in SYNC (metro) + ASYNC to DR; DR instance pre-installed, HANA stopped, storage kept current via replication
- **ASCS cluster** replicated; switch-over scripted via SAP toolset
- **Pilot-light for integrations:** middleware at DR in low-power mode, auto-scaled on failover trigger
- **Backup-restore** retained as Tier-3 fallback (24 h RTO) for catastrophic scenario

Why warm-standby meets RTO/RPO

- HANA in pre-attached mode: takeover $\leq$ 15 min
- ASCS + CI start sequence: 20 min
- Integration health & smoke tests: 30 min
- DNS cutover + user logon wave: 35 min
- **Total ~100 min**, with 20-min contingency inside 120 min target

Cost vs risk: ~40% of hot-standby cost; 5× faster than backup-restore.

ISO 22301 §8.3.

9. Recovery Procedure

ISO 22301:2019 Clause 8.4 — 165 words

Runbook — Phased Recovery

Phase 0 — Detect & Declare (T0 → T+15 min)

1. Monitoring alert fires (NOC) — auto-page IC, DR Coordinator
2. IC confirms outage, performs initial triage
3. If business impact = T1, IC declares **DISASTER** and authorises invocation
4. War-room opens; comms Lead drafts holding statement

Phase 1 — Failover (T+15 → T+90 min)

1. HANA SR takeover on DR site: `hdbnsutil -sr_takeover` (DBA Lead)
2. Verify DB consistency, start application services (ASCS, CI, Fiori)
3. Middleware auto-promotes DR adapters; integration queue replay begins
4. DNS failover: production CNAME → DR VIP (TTL 60 s)

Phase 2 — Verification (T+90 → T+120 min)

1. Smoke tests: FI posting, sales order, payroll interface, e-invoicing send
2. Integration health: 12/12 green
3. User wave logon: 95% in 60 min
4. IC declares **RECOVERED**

Phase 3 — Stand-Down (within 48 h)

1. Post-incident review, evidence pack to auditors
2. Decision: fail-back or operate at DR; if fail-back, schedule maintenance window

ISO 22301 §8.5.2.

10. Communication Plan

ISO 22301:2019 Clause 8.4.3 — 184 words

Stakeholder Map

Audience	Channel	Owner	Cadence
Executive / Crisis Team	War-room bridge + SMS	IC	At declare, every 30 min until recovered
3,000 internal users	Email, Teams, intranet banner	Comms Lead	At declare, at T+60, at recovery, post-mortem
Integration partners (12)	Direct call + email	Integration Lead	At declare, at integration green
Regulators (tax, data protection)	Formal letter	CFO + Legal	Within 24 h if data-loss / breach
Customers (order delays)	Status page + account mgr	Comms Lead	At declare, hourly until recovered
Vendors / hyperscaler	Account manager	DR Coordinator	At declare, daily during outage
Audit / Risk Committee	Post-event report	Plan Owner	Within 5 business days

Escalation Tree

NOC → IC (≤ 15 min) → CIO + CFO (≤ 30 min) → CEO (if > 4 h impact or regulatory).

Pre-approved templates for each audience held in comms vault.

ISO 22301 §7.4.

11. Validation and Acceptance Criteria

ISO 22301:2019 Clause 8.5 — 200 words

Acceptance Criteria (Pass/Fail Gates)

Gate	Test	Pass Threshold	Owner
G1	HANA DB consistency check	Zero errors	DBA Lead
G2	ASCS + CI services active	All green in MMC	Basis Lead
G3	12 satellite integrations	12/12 sending + receiving test msg	Integration Lead
G4	Smoke test — FI posting	Posts successfully to correct period	Functional Lead
G5	Smoke test — sales order	Order created, delivery, invoice triggered	SD Lead
G6	Smoke test — payroll file	Inbound file processed, no error	HCM Lead
G7	E-invoicing outbound	Accepted by tax gateway	SD Lead
G8	User logon rate	$\geq 95\%$ of 3,000 users in 60 min	IC
G9	Data integrity — last replicated log	$\Delta \leq 30$ min (RPO)	DBA Lead
G10	Performance — S/4 transaction response	$\leq$ baseline + 20%	Basis Lead

Any gate FAIL = exercise not passed; trigger root-cause and re-test.

ISO 22301 §8.5.3, §9.2.

12. Exercise and Testing Plan

ISO 22301:2019 Clause 8.5 — 174 words

Annual Test Schedule

Quarter	Type	Scope	Success Criteria
Q1	Tabletop	IC, DR Coord, Comms walk through declare → stand-down using scenario inject (ransomware on HANA)	All decisions made within 30 min of inject; gaps logged
Q2	Partial / Component	HANA SR takeover in isolated DR test instance; verify RTO 120, RPO 30	Takeover ≤ 15 min, last log ≤ 30 min old
Q3	**Full Failover**	Production-like DR invocation (impactful, planned); 4-h maintenance window	All 10 validation gates pass; real RTO ≤ 120 min, RPO ≤ 30 min
Q4	**Surprise**	DR Coordinator receives simulated call without notice; on-call team executes partial failover of one module (e.g., HCM)	Recovery of chosen module within its RTO within 2 h of surprise trigger

Exercise Governance

- Results logged in DR register; KPIs fed to performance section
- Failures generate corrective actions (see §14)
- Auditors invited annually to Q3

ISO 22301 §9.2.2.

13. Performance Evaluation

ISO 22301:2019 Clause 9 — 159 words

Key Performance Indicators

KPI	Target	Source
RTO achieved vs 120 min	≤ 100% (≤ 120 min)	Exercise + real-event log
RPO achieved vs 30 min	≤ 30 min data loss	Last replicated log timestamp
Validation gates passed	10/10 in Q3 full exercise	Exercise report
Mean time to declare	≤ 15 min from incident	IC log
Communication dispatch time	≤ 20 min from declare	Comms log
Exercise completion rate	4/4 per year	DR register
Plan currency	Reviewed within 90 days	Document mgmt.
Corrective actions closed on time	≥ 90%	CA tracker

Review Cadence

- **Monthly:** KPI snapshot to IT Risk
- **Quarterly:** trend report to Risk Committee with top 3 actions
- **Annually:** full plan review, BIA refresh, strategy reassessment

ISO 22301 §9.3.

14. Continual Improvement

ISO 22301:2019 Clause 10 — 161 words

Lessons Learned Process

1. Within 5 business days post-event/exercise, IC convenes post-incident review (PIR)
2. PIR captures: timeline, decisions, gaps, near-misses, stakeholder feedback
3. Findings logged in DR improvement register with owner, severity, due date

Corrective Action Workflow

Step	Owner	Output
1. Identify finding	PIR participants	Logged in register
2. Root-cause (5-Whys / fishbone)	DR Coordinator	RCA doc
3. Action plan + owner + due date	Plan Owner	CA ticket
4. Implementation	Action owner	Evidence (config, script, training)
5. Verification	DR Coordinator	Re-test / sign-off
6. Closure	Plan Owner	Audit trail

Version Update Rule

- **Minor ($\leq 2 \times/\text{yr}$):** contact, tool, vendor changes
- **Major (annual or post-major incident):** strategy, BIA, RTO/RPO, structural changes
- All changes versioned in plan header, communicated to stakeholders, change-log retained

ISO 22301 §10.1, §10.2 — Continual Improvement.

Document Control

Field	Value
Document Type	Disaster Recovery Plan
Standard	ISO 22301:2019 — Security and resilience — Business continuity management systems
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